

Effect of calabash chalk on the histomorphology of the gastro-oesophageal tract of growing wistar rats.

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Calabash chalk is a naturally occurring mineral consumed by members of some Nigerian communities for pleasure and by pregnant women as a remedy for morning sickness. The consumption of this geophagic material motivated our interest on the effect of the chalk on the histomorphology of the gastro-oesophageal tract. Twenty-eight young Wistar rats, 4 weeks old, were divided into 4 groups of equal size. Group 1 animals served as controls and received 1 mL of distilled water. Groups 2, 3, and 4 received orally 1 mL of a Calabash chalk suspension containing 40 mg/mL for 14, 21, and 28 days, respectively. Upon completion of the treatments, the animals in groups 2, 3, and 4 were sacrificed on days 15, 22, and 29, respectively, and the control group animals were sacrificed on day 29. All animals were euthanised using chloroform anaesthesia. The oesophagus and the stomach of each animal were dissected out and routinely processed for histological studies. There was oedema with haemorrhages in the mucosa of the stomach, and acanthosis, hyperkeratosis, and koilocytic changes were observed in the mucosa of the oesophagus of the groups treated with 40 mg/mL of Calabash chalk suspension. Calabash chalk caused histological changes to the stomach and the oesophagus that may lead to other pathophysiological conditions.

Evaluation of prenatal calabash chalk geophagy on the developing brain of Wistar rats.

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Calabash chalk (CaC) is an aluminium silicate hydroxide compound with heavy metal constituents, making it a potential neurotoxicant. Pregnant women often consume CaC as an antiemetic, which may interfere with the normal development of the foetal brain. Here, we evaluated the effects of CaC administration in pregnant rats on the brain of the offspring. Wistar rat dams were assigned to one of three groups: control, 200 mg/kg and 800 mg/kg of a CaC suspension. Administrations lasted 14 days (gestation days 7-20). On day 14, 5-bromo-2'-deoxyuridine (BrdU) was administered and dams were allowed to term. Behavioural tests were performed on different days as the pups matured, and they were sacrificed on post-natal days 30 and 60. Brains were processed for histology and Western blotting. Results showed no significant differences in surface righting reflex, cliff avoidance, negative geotaxis and open-field activity. No hippocampal and somatosensory cortical cytoarchitectonic alterations and no significant signs of glial fibrillary acidic protein (GFAP) activation were observed. Neuronal nuclei counts showed variability in the somatosensory cortex and hippocampus of the CaC group. BrdU-positive cells were significantly lower in the 200 mg/kg group and higher in the 800 mg/kg group. Doublecortin-X-positive cells were not different in all the CaC groups. Astrocytes and microglia Western blotting quantification confirmed no significant increase in pup glial cells in adulthood. Prenatal consumption of CaC at indicated dosages may not be deleterious to the developing brain, especially after cessation of exposure and during maturation of the animal. However, the differences in neuronal and glial populations may be due to their ability to cope with CaC.

High umbilical cord blood lead levels and "calabar chalk" consumption amongst pregnant women in two hospitals in Cameroon.

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"Calabar chalk" is a lead-laden pica mostly consumed by pregnant women worldwide as a remedy for morning sickness. This clay material has been shown to have lead levels of up to 40mg per kilogram. Meanwhile blood lead levels, even at doses less than 10µg/dl will be toxic to humans and even worse-off to the fetus as it crosses the placenta. We, therefore, sort to determine the prevalence of "Calabar chalk" consumption amongst pregnant women and if it translates to higher umbilical cord blood lead levels. We carried out a cross-sectional study by prospectively and consecutively enrolling 300 pregnant women from December 2014 through February 2015. A questionnaire was administered to ascertain "Calabar chalk" consumption. The levels of lead in the umbilical cord blood of 51 participants of each group of those who consumed and didn't consume "Calabar chalk" were measured by spectrometry and compared using the T-test ($p < 0.05$). The prevalence of "Calabar chalk" consumption was 43.33%. This was mostly consumed during pregnancy only (46.34%), with higher rates observed amongst primigravidas. The mean umbilical blood lead levels amongst those who consumed and those who did not consume "Calabar chalk" was 39.19µg/dl and 25.33µg/dl respectively ($P = 0.111$). The prevalence of "Calabar chalk" consumption was high in the pregnant women population. The overall umbilical cord blood lead levels were extremely high in both consumers and non-consumers. We recommend health education and chelation therapy to be considered.

Comparative Efficacy and Safety of Glucagon-like Peptide-1 Receptor Agonists in Children and Adolescents with Obesity or Overweight: A Systematic Review and Network Meta-Analysis.

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Four glucagon-like peptide-1 receptor agonists (GLP-1 RAs) have been used in children and adolescents with obesity or overweight. This network meta-analysis was conducted to compare the efficacy and safety of these regimens. Embase, PubMed, and Scopus were searched on March 2023 and updated in June 2024 for eligible randomized controlled trials (RCTs). The primary efficacy outcomes were mean difference in actual body weight, BMI (body mass index), BMI z score, and waist circumference. Safety outcomes included nausea, vomiting, diarrhea, abdominal pain, injection-site reaction, and hypoglycemia. Eleven RCTs with 953 participants were eligible. Semaglutide exhibited greater effects in reducing weight, BMI, and BMI z score versus the placebo. Semaglutide was associated with greater weight loss and BMI z score reduction in comparison with exenatide, liraglutide, and dulaglutide. Semaglutide also significantly decreased BMI than exenatide. None of the four GLP-1 RAs were associated with higher risks of diarrhea, headache, and abdominal pain versus the placebo. Liraglutide was more likely to cause nausea, vomiting, hypoglycemia, and injection-site reactions than the placebo. Liraglutide also had higher odds of causing injection-site reactions than other GLP-1 RAs. Semaglutide appeared to be the most effective and safe option among four GLP-1 RAs in children and adolescents with obesity or overweight.

Glucagon-like-peptide 1 receptor agonism and attempted suicide: A Mendelian randomisation study to assess a potential causal association.

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Glucagon-like-peptide 1 receptor agonists (GLP-1RA) have transformed type 2 diabetes (T2D) and obesity management. Multiple regulatory agencies are investigating reported associations between GLP1-RA and increased suicide attempts (SA), but observational data may be prone to confounding. Randomised control trials (RCT) of GLP-1RA were largely undertaken in people at lower risk of SA. Real-world data suggest semaglutide use associates with reduced suicidal ideation and depression but was under-powered to statistically assess risk of SA. Mendelian randomisation (MR) leverages genetic instrument(s) to infer potential causal association between an exposure and an outcome. We undertook MR using missense variants in the gene encoding GLP1R that improve glycemia, lower T2D risk and/or lower BMI, to investigate potential causal association between GLP-1RA and SA. In people of European ancestry, MR did not find evidence genetically proxied GLP1RA increased SA in a general population cohort: (rs10305492, exposure: HbA1c, odds ratio [OR] and 95% confidence interval [CI]: 1.38, 0.41-4.62, $p = .60$), (rs10305492, exposure: FG, OR 1.27, 0.52-3.13, $p = .60$) and (rs1042044, exposure BMI, OR 0.30, 0.06-1.48) with concordant results in a multi-ancestry SA case-control cohort. In conclusion, we did not find MR evidence that increased GLP-1RA impacts SA. This awaits confirmation with RCT and real-world data.

Practice Patterns of Glucagon-Like Peptide-1 Agonist Use Among Pediatric Bariatric Surgeons: A National Survey.

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Glucagon-like peptide-1 receptor agonist (GLP-1A) medications are gaining widespread popularity for the treatment of obesity. The optimal use of these drugs in pediatric bariatric populations, and especially in those considering metabolic and bariatric surgery (MBS), is yet to be established. We sought to characterize current practice patterns of GLP-1A use at major pediatric bariatric centers across the United States. We administered an online survey to a purposive sample of 46 surgeons who perform MBS on children and adolescents. Survey questions explored practices prescribing GLP-1As in patients considering MBS, holding them prior to elective operations, and restarting them postoperatively following MBS. Responses were summarized with descriptive statistics and inductive content analysis. There were 22 responses (48% response rate) representing 19 institutions. Most (86%) respondents do sometimes prescribe GLP-1As for patients considering MBS, but the specific indications vary. Practices for holding GLP-1As preoperatively also vary, from not at all to holding for 2 wk. Over half (55%) of respondents sometimes restart GLP-1As after MBS. Free-response themes included still-evolving preoperative utilization patterns, difficulty with access and insurance coverage, and a lack of data informing GLP-1A use in the pre and postoperative periods. Given the increasing use of these medications for weight loss purposes, this substantial variation in practice highlights a need for further research to examine the safest and most effective use of GLP-1As in the pre and postoperative periods and for practice guidelines to standardize care pathways in pediatric bariatric contexts.

The effects of incretin-based therapies on weight reduction and metabolic parameters in children with obesity: A systematic review and meta-analysis.

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Growing evidence indicates that incretin-based therapies (IBTs), glucagon-like peptide-1 receptor

agonists (GLP-1 RAs), and dipeptidyl peptidase-4 inhibitors (DPP4is) are effective and safe for treating pediatric obesity patients with or without type 2 diabetes. Therefore, we aimed to perform a systematic review and meta-analysis for updating current evidence. We searched the PubMed, the Cochrane Library, and the EMBASE database for articles published until September 15, 2023, and limited to randomized control trials. The primary outcomes were changed from baseline in weight metrics and the cardiometabolic profile. A random effects model will be used, as high heterogeneity is expected. All analyses were performed using STATA 17.0. Fifteen trials with a total number of 1286 participants were included in our meta-analysis. Overall, the mean difference in weight change between the IBTs group and the control group was -2.89 kg (95% confidence interval, -5.12 to -0.65, $p = 0.011$). Additionally, IBTs significantly reduced the HbA1c level and fasting plasma glucose by 0.37% and 6.99 mg/dl, compared with control groups. IBTs showed a little increased risk of GI side effects and hypoglycemia events, but none of the severe hypoglycemia events were occurred in IBTs group. Our study results have proved that GLP-1 RAs are safe, acceptable, and effective in weight reduction and sugar control for children with obesity. In addition, DPP-4is seems to have no effect on glycemic control and weight loss in childhood obesity. Further research is needed to confirm these findings, especially in younger children.

Use of GLP-1 Receptor Agonists for the Management of Type 1 Diabetes: A Pediatric Perspective.

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Despite all the technological advances in treatment of patients with type 1 diabetes (T1D), glucose control remains suboptimal in most patients. In addition, a relatively high percentage of patients with T1D, including children, have obesity. Therefore, new interventions are required that focus their effects on weight loss, in order to help with associated insulin resistance and improve glycemic control. GLP-1 receptor agonists (GLP-1 RAs) have proven to be effective and safe in adults with T1D, showing improvement in glycemic control, body weight and cardiorenal protection. GLP-1 RAs are also approved for children with obesity (above the age of 12 years) or type 2 diabetes (above the age of 10 years). However, currently these medications are not approved for use in children with T1D. Only a few published studies have evaluated their efficacy and safety for this indication. This review presents the rationale and experience of add-on GLP-1 RA therapy to pediatric and adolescent patients with T1D, otherwise treated, from RCTs and real-world data. Results of studies of GLP-1 RA in children with T1D are still pending, while large multicenter randomized controlled trials (RCTs) in this population are lacking.

Update on Pediatric Anti-obesity Medications-Current Landscape and Approach to Prescribing.

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To review the current medical therapies available for treatment of obesity in children and adolescents less than 18 years old in the United States and outline the approach to their use. Obesity is a chronic disease with increasing prevalence in children and adolescents in the United States. Over the past few years, more FDA-approved medical treatments for obesity, such as GLP-1 receptor agonists, have emerged for patients less than 18 years old. Furthermore, there are medications with weight loss effects that can be used off-label for obesity in pediatric patients. However, access to many of these medications

is limited due to age restrictions, insurance coverage, and cost. Medical options are improving to provide treatment for obesity in pediatric populations. FDA and off-label medications should be considered when appropriate to treat children and adolescents with obesity. However, further studies are needed to evaluate the efficacy and long-term safety of FDA-approved and off-label medications for obesity treatment in pediatric patients.

Editorial: Potentials and Pitfalls in Targeting Glucagon-Like Peptide-1 (GLP-1) in the Management of Increasing Levels of Obesity.

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Rising levels of obesity in all age groups are associated with profound effects on health and economies in developed and developing countries. This year, the scientific research behind the development of glucagon-like peptide 1 (GLP-1) receptor agonists (GLP-1 analogs or incretin mimetics) has been recognized. On 19 September 2024, three scientists were presented with a Lasker Award for their basic clinical research on identifying and studying the roles of GLP-1. The research by Joel Habener, Svetlana Mojsov, and Lotte Bjerre Knudsen began three decades ago and has led to new anti-obesity drugs, which mimic GLP-1 to lower blood glucose levels and control appetite. The efficacy of GLP-1 receptor agonists in the management of obesity in adults, as well as in children and adolescents, has now been supported by several clinical trials. This editorial aims to describe the research behind developing GLP-1 receptor agonists and their potential and pitfalls in managing obesity in all age groups.
